

In Exercises 35–42, use the laws in Definition 1 to show that the stated properties hold in every Boolean algebra.

35. Show that in a Boolean algebra, the **idempotent laws** $x \vee x = x$ and $x \wedge x = x$ hold for every element x .
 36. Show that in a Boolean algebra, every element x has a unique complement $\bar{x}$ such that $x \vee \bar{x} = 1$ and $x \wedge \bar{x} = 0$.
 37. Show that in a Boolean algebra, the complement of the element 0 is the element 1 and vice versa.
 38. Prove that in a Boolean algebra, the **law of the double complement** holds; that is, $\overline{\bar{x}} = x$ for every element x .
 39. Show that **De Morgan's laws** hold in a Boolean algebra.
- That is, show that for all x and y , $\overline{(x \vee y)} = \bar{x} \wedge \bar{y}$ and $\overline{(x \wedge y)} = \bar{x} \vee \bar{y}$.
40. Show that in a Boolean algebra, the **modular properties** hold. That is, show that $x \wedge (y \vee (x \wedge z)) = (x \wedge y) \vee (x \wedge z)$ and $x \vee (y \wedge (x \vee z)) = (x \vee y) \wedge (x \vee z)$.
 41. Show that in a Boolean algebra, if $x \vee y = 0$, then $x = 0$ and $y = 0$, and that if $x \wedge y = 1$, then $x = 1$ and $y = 1$.
 42. Show that in a Boolean algebra, the **dual** of an identity, obtained by interchanging the $\vee$ and $\wedge$ operators and interchanging the elements 0 and 1, is also a valid identity.
 43. Show that a complemented, distributive lattice is a Boolean algebra.

12.2 Representing Boolean Functions

Two important problems of Boolean algebra will be studied in this section. The first problem is: Given the values of a Boolean function, how can a Boolean expression that represents this function be found? This problem will be solved by showing that any Boolean function can be represented by a Boolean sum of Boolean products of the variables and their complements. The solution of this problem shows that every Boolean function can be represented using the three Boolean operators $\cdot$, $+$, and $\bar{}$. The second problem is: Is there a smaller set of operators that can be used to represent all Boolean functions? We will answer this question by showing that all Boolean functions can be represented using only one operator. Both of these problems have practical importance in circuit design.

12.2.1 Sum-of-Products Expansions

We will use examples to illustrate one important way to find a Boolean expression that represents a Boolean function.

EXAMPLE 1 Find Boolean expressions that represent the functions $F(x, y, z)$ and $G(x, y, z)$, which are given in Table 1.

TABLE 1					
x	y	z	F	G	
1	1	1	0	0	
1	1	0	0	1	
1	0	1	1	0	
1	0	0	0	0	
0	1	1	0	0	
0	1	0	0	1	
0	0	1	0	0	
0	0	0	0	0	

Solution: An expression that has the value 1 when $x = z = 1$ and $y = 0$, and the value 0 otherwise, is needed to represent F . Such an expression can be formed by taking the Boolean product of x , $\bar{y}$, and z . This product, $x\bar{y}z$, has the value 1 if and only if $x = \bar{y} = z = 1$, which holds if and only if $x = z = 1$ and $y = 0$.

To represent G , we need an expression that equals 1 when $x = y = 1$ and $z = 0$, or $x = z = 0$ and $y = 1$. We can form an expression with these values by taking the Boolean sum of two different Boolean products. The Boolean product $xy\bar{z}$ has the value 1 if and only if $x = y = 1$ and $z = 0$. Similarly, the product $\bar{x}\bar{y}z$ has the value 1 if and only if $x = z = 0$ and $y = 1$. The Boolean sum of these two products, $xy\bar{z} + \bar{x}\bar{y}z$, represents G , because it has the value 1 if and only if $x = y = 1$ and $z = 0$, or $x = z = 0$ and $y = 1$. 

Example 1 illustrates a procedure for constructing a Boolean expression representing a function with given values. Each combination of values of the variables for which the function has the value 1 leads to a Boolean product of the variables or their complements.

Definition 1

A *literal* is a Boolean variable or its complement. A *minterm* of the Boolean variables $x_1, x_2, \dots, x_n$ is a Boolean product $y_1 y_2 \cdots y_n$, where $y_i = x_i$ or $y_i = \bar{x}_i$. Hence, a minterm is a product of n literals, with one literal for each variable.

A minterm has the value 1 for one and only one combination of values of its variables. More precisely, the minterm $y_1 y_2 \cdots y_n$ is 1 if and only if each y_i is 1, and this occurs if and only if $x_i = 1$ when $y_i = x_i$ and $x_i = 0$ when $y_i = \bar{x}_i$.

EXAMPLE 2 Find a minterm that equals 1 if $x_1 = x_3 = 0$ and $x_2 = x_4 = x_5 = 1$, and equals 0 otherwise.

Solution: The minterm $\bar{x}_1 x_2 \bar{x}_3 x_4 x_5$ has the correct set of values. 

Links

By taking Boolean sums of distinct minterms we can build up a Boolean expression with a specified set of values. In particular, a Boolean sum of minterms has the value 1 when exactly one of the minterms in the sum has the value 1. It has the value 0 for all other combinations of values of the variables. Consequently, given a Boolean function, a Boolean sum of minterms can be formed that has the value 1 when this Boolean function has the value 1, and has the value 0 when the function has the value 0. The minterms in this Boolean sum correspond to those combinations of values for which the function has the value 1. The sum of minterms that represents the function is called the **sum-of-products expansion** or the **disjunctive normal form** of the Boolean function.

(See Exercise 46 in Section 1.3 for the development of disjunctive normal form in propositional calculus.)

EXAMPLE 3 Find the sum-of-products expansion for the function $F(x, y, z) = (x + y)\bar{z}$.

Extra Examples

Solution: We will find the sum-of-products expansion of $F(x, y, z)$ in two ways. First, we will use Boolean identities to expand the product and simplify. We find that

$$\begin{aligned}
 F(x, y, z) &= (x + y)\bar{z} \\
 &= x\bar{z} + y\bar{z} && \text{Distributive law} \\
 &= x1\bar{z} + 1y\bar{z} && \text{Identity law} \\
 &= x(y + \bar{y})\bar{z} + (x + \bar{x})y\bar{z} && \text{Unit property} \\
 &= xy\bar{z} + x\bar{y}\bar{z} + xy\bar{z} + \bar{x}y\bar{z} && \text{Distributive law} \\
 &= xy\bar{z} + x\bar{y}\bar{z} + \bar{x}y\bar{z}. && \text{Idempotent law}
 \end{aligned}$$

Second, we can construct the sum-of-products expansion by determining the values of F for all possible values of the variables x , y , and z . These values are found in Table 2. The sum-of-products expansion of F is the Boolean sum of three minterms corresponding to the three rows of this table that give the value 1 for the function. This gives

$$F(x, y, z) = xy\bar{z} + x\bar{y}\bar{z} + \bar{x}y\bar{z}. \quad \text{img alt="blue triangle pointing right" data-bbox="870 810 885 825"/>$$

It is also possible to find a Boolean expression that represents a Boolean function by taking a Boolean product of Boolean sums. The resulting expansion is called the **conjunctive normal form** or **product-of-sums expansion** of the function. These expansions can be found from sum-of-products expansions by taking duals. How to find such expansions directly is described in Exercise 10.

TABLE 2					
x	y	z	$x + y$	$\bar{z}$	$(x + y)\bar{z}$
1	1	1	1	0	0
1	1	0	1	1	1
1	0	1	1	0	0
1	0	0	1	1	1
0	1	1	1	0	0
0	1	0	1	1	1
0	0	1	0	0	0
0	0	0	0	1	0

12.2.2 Functional Completeness

Every Boolean function can be expressed as a Boolean sum of minterms. Each minterm is the Boolean product of Boolean variables or their complements. This shows that every Boolean function can be represented using the Boolean operators $\cdot$, $+$, and $\bar{}$. Because every Boolean function can be represented using these operators we say that the set $\{\cdot, +, \bar{}\}$ is **functionally complete**. Can we find a smaller set of functionally complete operators? We can do so if one of the three operators of this set can be expressed in terms of the other two. This can be done using one of De Morgan's laws. We can eliminate all Boolean sums using the identity

$$x + y = \overline{\bar{x}\bar{y}},$$

which is obtained by taking complements of both sides in the second De Morgan law, given in Table 5 in Section 12.1, and then applying the double complementation law. This means that the set $\{\cdot, \bar{}\}$ is functionally complete. Similarly, we could eliminate all Boolean products using the identity

$$xy = \overline{\bar{x} + \bar{y}},$$

which is obtained by taking complements of both sides in the first De Morgan law, given in Table 5 in Section 12.1, and then applying the double complementation law. Consequently $\{+, \bar{}\}$ is functionally complete. Note that the set $\{+, \cdot\}$ is not functionally complete, because it is impossible to express the Boolean function $F(x) = \bar{x}$ using these operators (see Exercise 19).



We have found sets containing two operators that are functionally complete. Can we find a smaller set of functionally complete operators, namely, a set containing just one operator? Such sets exist. Define two operators, the $|$ or **NAND** operator, defined by $1 | 1 = 0$ and $1 | 0 = 0 | 1 = 0 | 0 = 1$; and the $\downarrow$ or **NOR** operator, defined by $1 \downarrow 1 = 1 \downarrow 0 = 0 \downarrow 1 = 0$ and $0 \downarrow 0 = 1$. Both of the sets $\{| \}$ and $\{\downarrow\}$ are functionally complete. To see that $\{| \}$ is functionally complete, because $\{\cdot, \bar{}\}$ is functionally complete, all that we have to do is show that both of the operators $\cdot$ and $\bar{}$ can be expressed using just the $|$ operator. This can be done as

$$\begin{aligned}\bar{x} &= x | x, \\ xy &= (x | y) | (x | y).\end{aligned}$$

The reader should verify these identities (see Exercise 14). We leave the demonstration that $\{\downarrow\}$ is functionally complete for the reader (see Exercises 15 and 16).